Doc

J LP

March 30, 2026

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1 Raw text

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit

ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

2 Basics - sections, subsections, paragraphs...

2.1 Normal

Hello Normal

2.1.1 Normal

A paragraph yeah a paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens.

2.1.2 Normal2

Comon we do it again A paragraph yeah a paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens.

Important paragraph A paragraph yeah a paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens. Comon we do it again A paragraph yeah a paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens.

Important sub-paragraph A sub-paragraph yeah a sub-paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens. Comon we do it again A paragraph yeah a paragraph haha, lol i dk what to write here so i rpetend just to write sentences that make sens whil it is pure nonsens.

Hello in bold

Hello in italic

Hello underlined

Hello all

3 Font Sizes

tiny

footnotesize

small

normalsize

large

Large

huge

Huge

4 Footnotes

Hell0[1]

Hell02[2]

5 Special characters you must escape with \

$

%

$

—

6 Links

My-blog

https://julienlargetpiet.tech

https://julienlargetpiet.tech

My-blog

[1] this is a first footnote
[2] this is a second footnote

7 Xcolors

Important

Important

Important

8 Highligts

Default coloredbox

This is a highlighted block

Customoized tcolorbox

This is a second highlighted block

Customoized 2 tcolorbox

This is a second highlighted block

Customoized 3 tcolorbox

Hell Yeah Title

Clean modern box

9 Math

Some Math:

$$\int_0^1 x^2 dx = \frac{1}{3}$$

Inline maths:

$\int_0^1 x^2 dx = \frac{1}{3}$

10 Code-Block

First, with verbatim

```
int main() {
    return 0;
}
```

Now with listings

Variable version settings:

```
#include <iostream.h>

int main() {
    std::string x = "this is a string";
    std::cout << x << "\n";
    // comment
    return 0;
}
```

With minted (best imo):

```
#include <iostream.h>

int main() {
    std::string x = "this is a string";
    std::cout << x << "\n";
    // comment
    return 0;
}
```

11 Lists

- First
- Second

Numbered Lists:

1. First
2. Second

12 Tables

A	B
C	D

13 Images

Figure 1: Example Image

14 Expansions

14.1 Variables (Macro basiques)

Bonjour Rémi et Julien.

Bonjour Rémi et Julien.

14.2 Macro - Extended

Hello Rémi, Julien et Lucas

14.3 Les conditions

OUI_1

NO_2

NO_2B

YES

NO_3

OUI_4

OUI_5

OUI_6

NO_6

OUI_7

OUI_8

NO_9

14.4 Check Emptyness

EMPTY

14.5 Loop

Value: 1
Value: 2
Value: 3
END

Value: 1
Value: 2
Value: 3
END

A: 1
B: 2
C: 3
D: 4

A: 1: A1
B: 2: B2
C: 3: C3
D: 4: D4

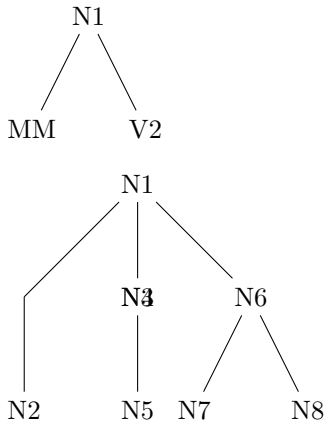
A: 1 - A1
B: 2 - B2
C: 3 - C3
D: 4 - D4

A: 1 - A1
B: 2 - B2
C: 3 - C3
D: 4 - D4

14.6 Expandre un arbre

N1: Side: L/V1
N2: Side: R/V2
N3: (Side: L/Value: V3)/R/V4
N4: (Side: L/((Side: L/Value: V5)/(Side: R/Value: V6)))/R/V7

14.7 Tikz - Trees



14.7.1 L'arbre prend forme

En fait j'ai pas réusii, il vaut mieux générer la structure statique avec python, R ou autre puis le mettre dans un include/input.

15 Cool macro

OUI

16 Boxes

Hello	Hello	Hello

17 Includes, yess like C

17.1 Input - no newpage

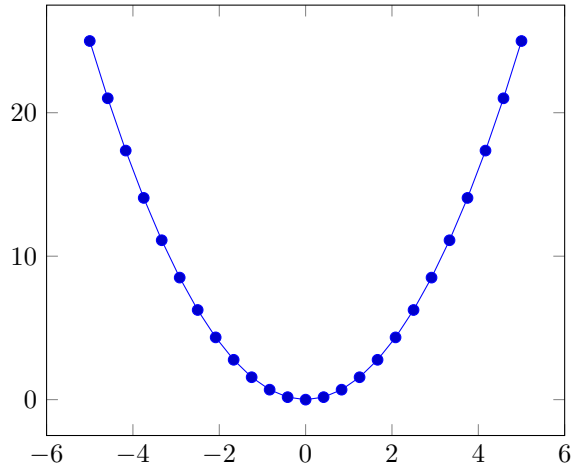
Je suis inclu

Je suis inclu - 2

17.2 Real incudes

Je suis inclu

18 Tikz



19 MultiColumn - Nice for just text

19.1 Manual column breaking

This is col 1.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

This is col 2.

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19.2 Auto column breaking

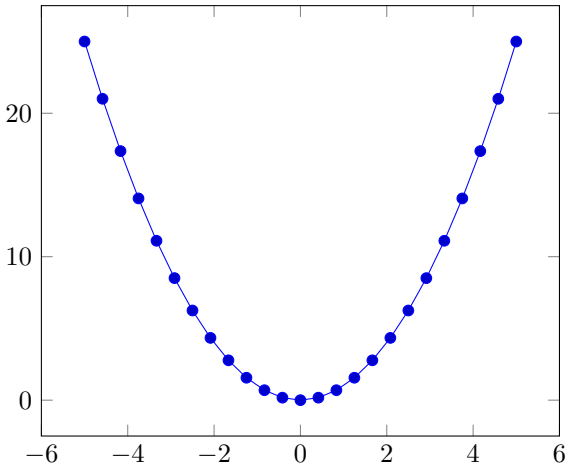
Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

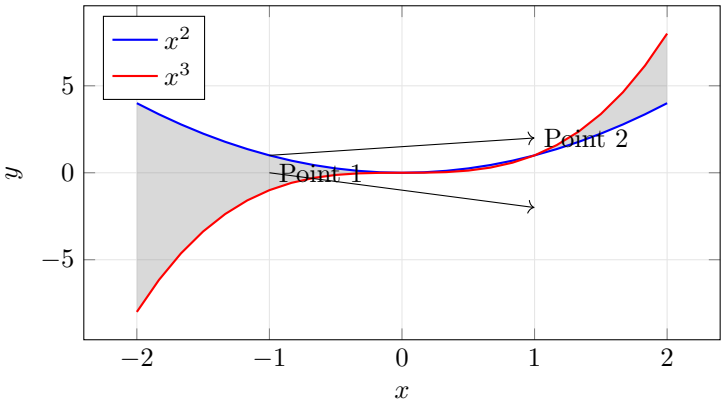
19.3 Minipage - You can use Tikz here for each column

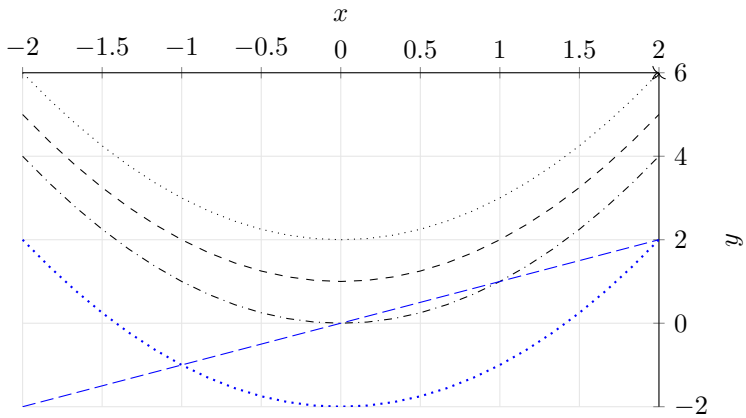
Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

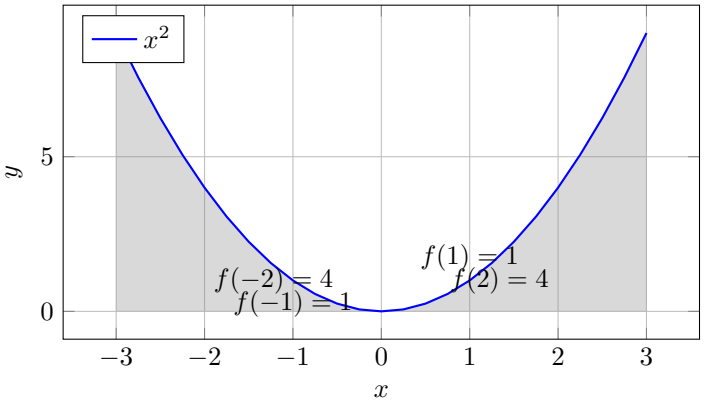


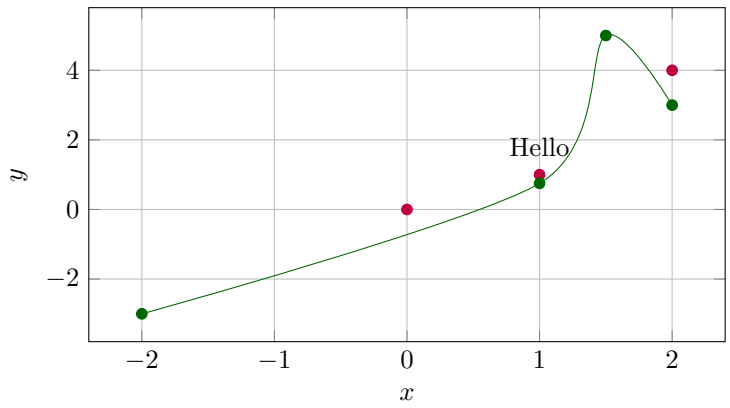
20 More on Tikz - pgfplots

20.1 2D graphs

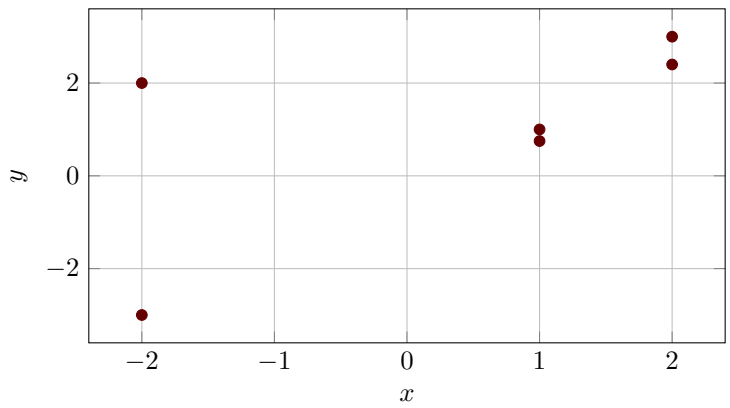




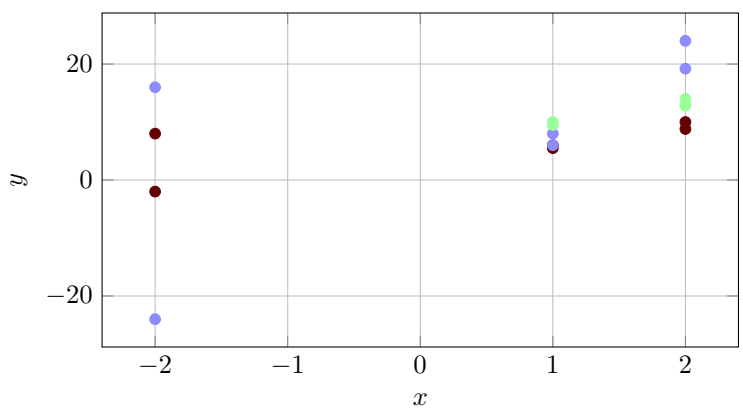




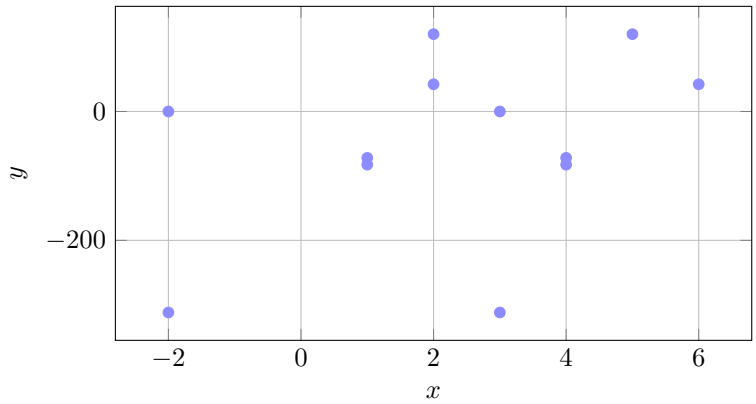
20.1.1 Scatterplot - CSV



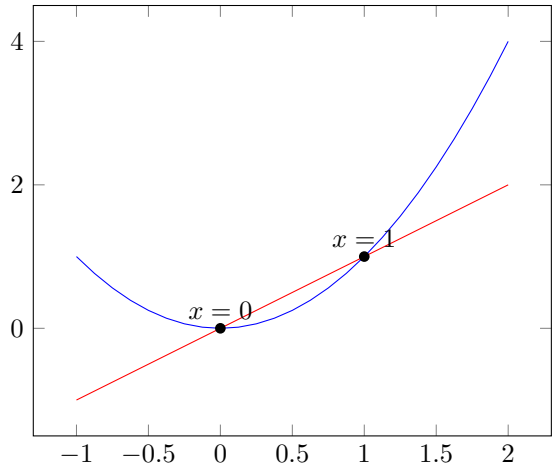
20.1.2 Scatterplot - CSV - Operations inside Table



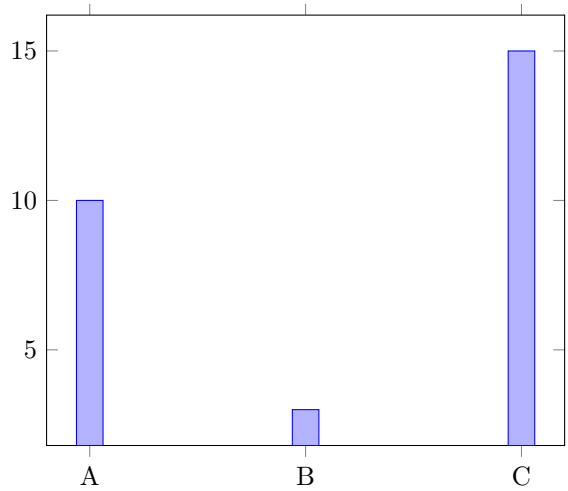
20.1.3 Scatterplot - CSV - Store in memory + Vertical concat



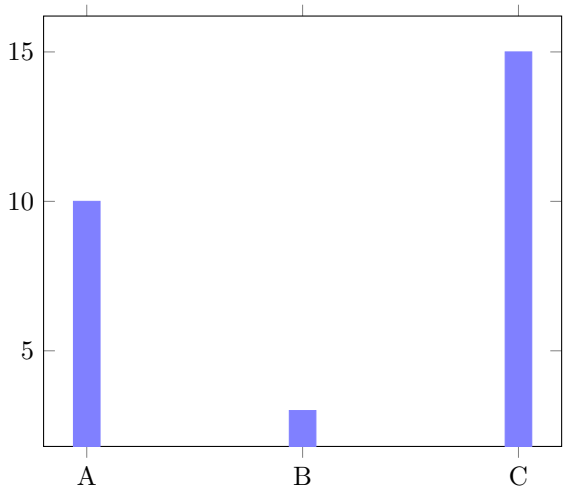
20.1.4 Intersections



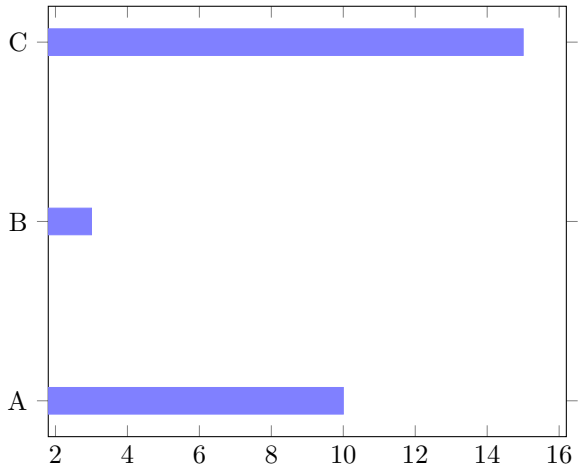
20.2 Barplot



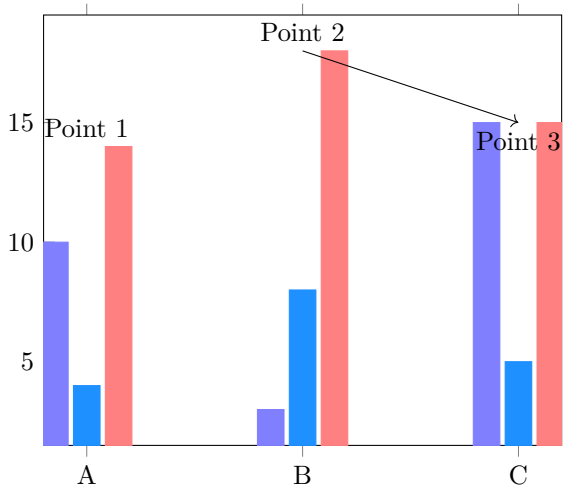
20.3 With CSV



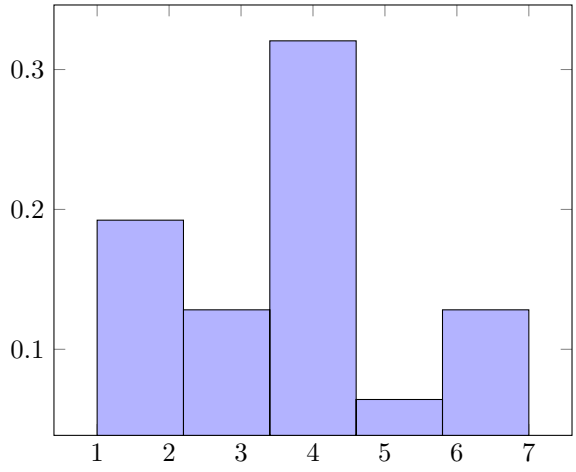
20.4 Rotating barplots



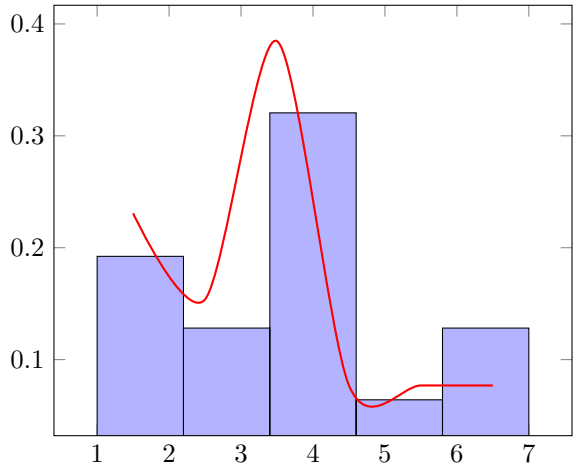
20.4.1 Grouped Barplot



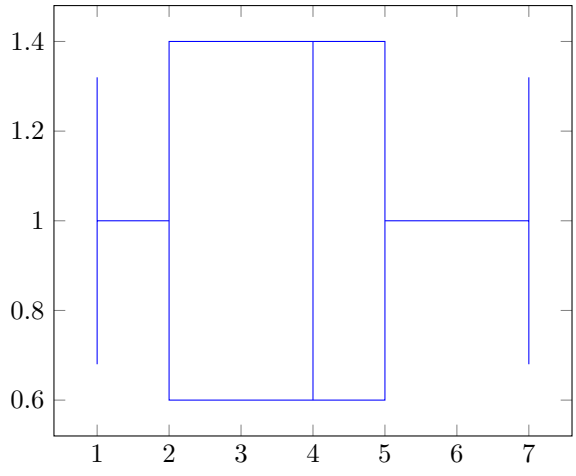
20.5 Histogramms + CSV



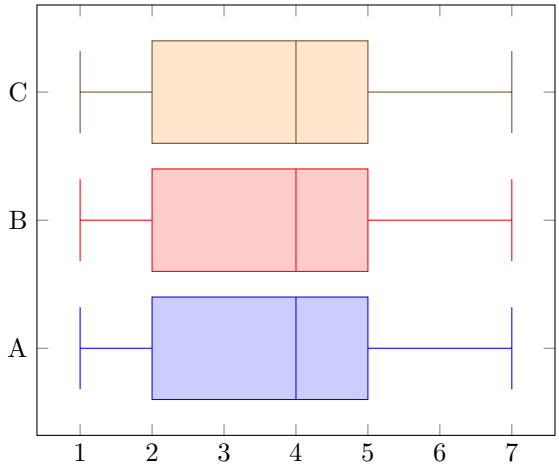
20.6 Adding KDE plot



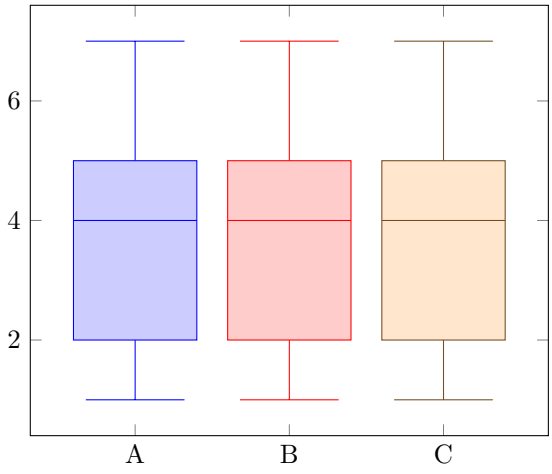
20.7 Horizontal Boxplots + CSV



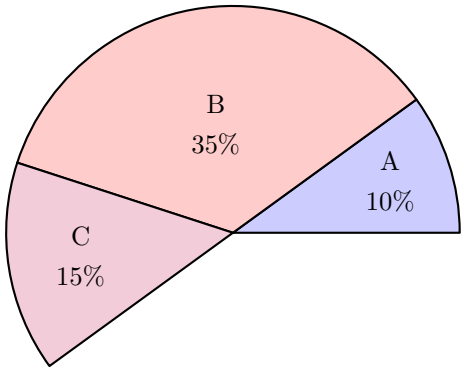
20.8 Multiple histogramms + CSV



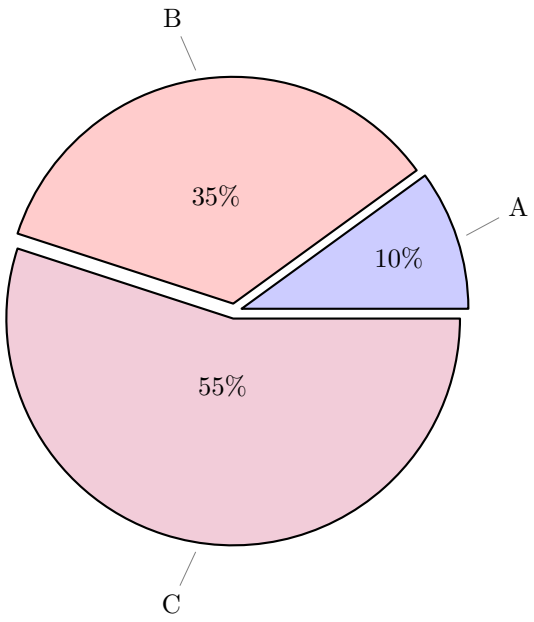
20.9 Vertical Boxplots + CSV



20.10 Pie Chart - Incomplete



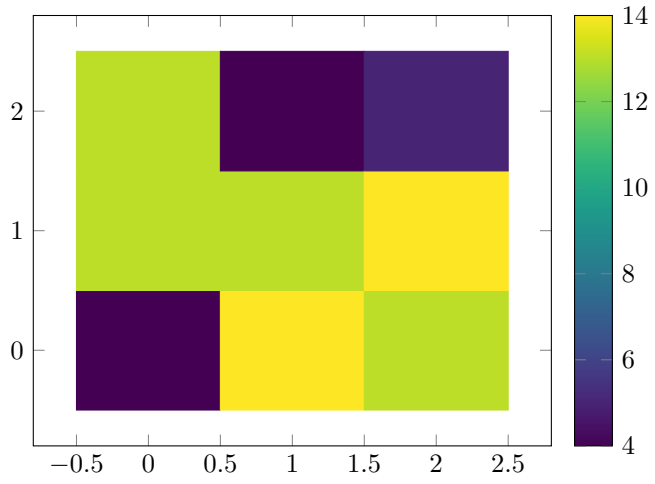
20.10.1 Full Circle



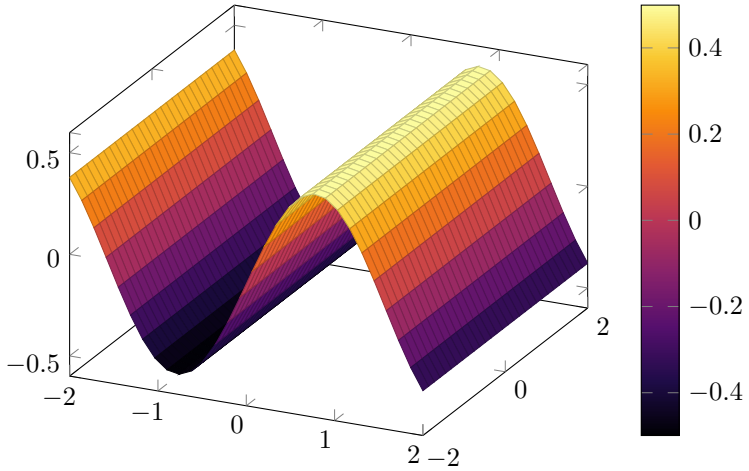
20.10.2 With CSV

20.11 HeatMaps

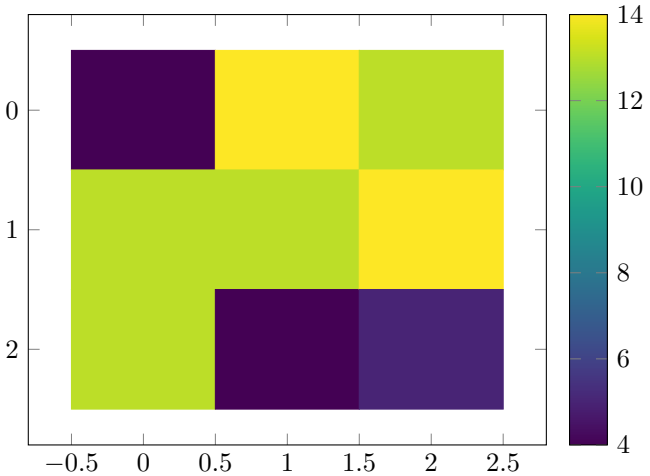
20.11.1 Color interpolation



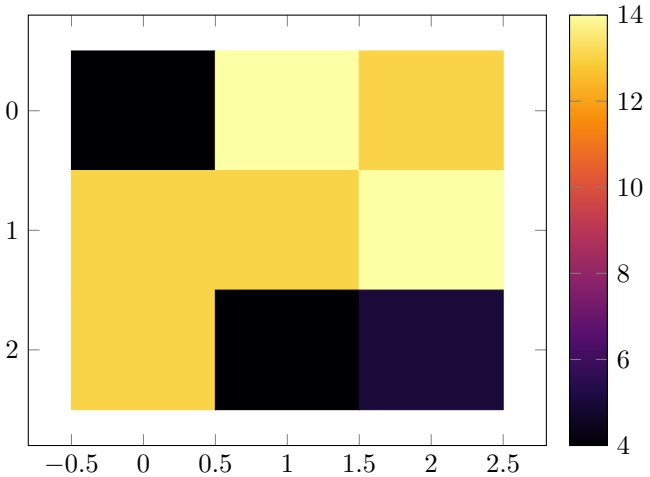
21 3D graphs



21.0.1 Fixed color

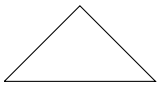


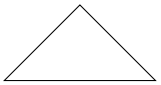
21.0.2 CSV



22 Tikz - Shapes

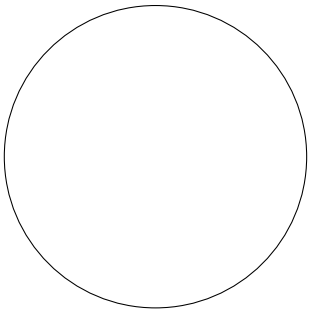
22.1 Triangle



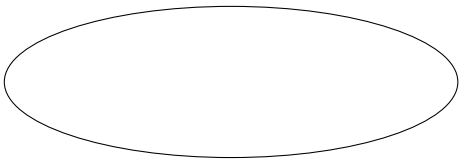


22.2 Rectangle

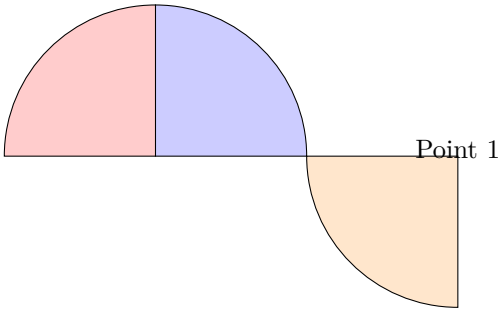
22.3 Cirlce



22.4 Elips

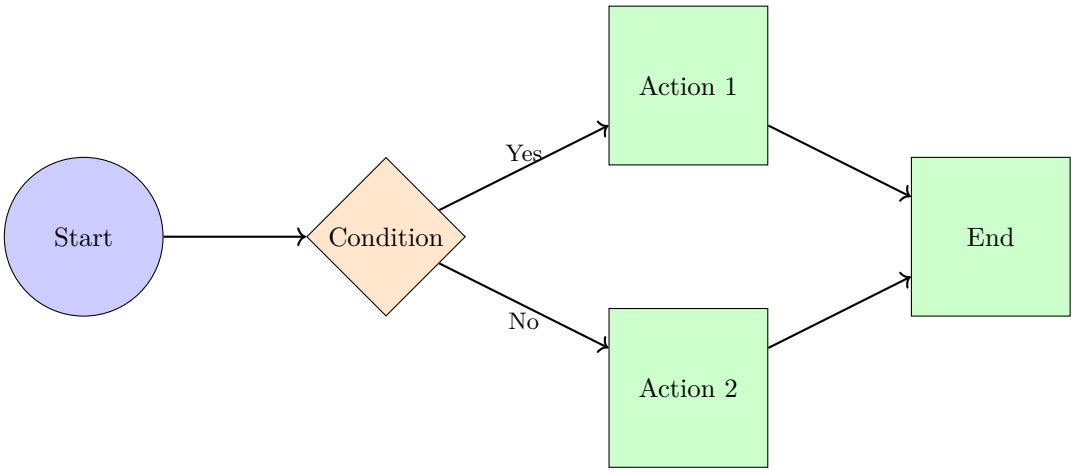


22.5 Arcs

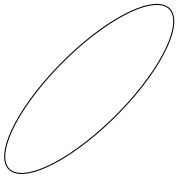


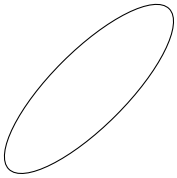
22.6 Nodes & Shapes - Connectons

Another example:

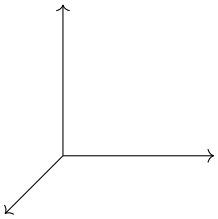


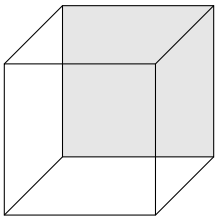
22.7 Transformations





22.8 3D





22.8.1 Camera angles

Here we deonstrate the importance of rotation composition order.

If we srite:

rotate around x rotate around y rotate around z

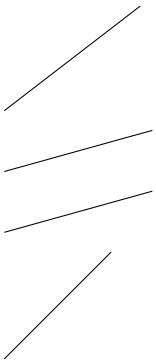
It will be this composition:

rotate_x(rotate_y(rotate_z()))

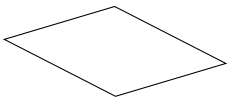
Rotating a line that is perfectly aligned to axis X around X changes nothing (logic).

So in that is why second and third figures are the same, because in the third figure, rotating around x happens first and does nothing, so still jut y rotation that is the same at what the second figure does.

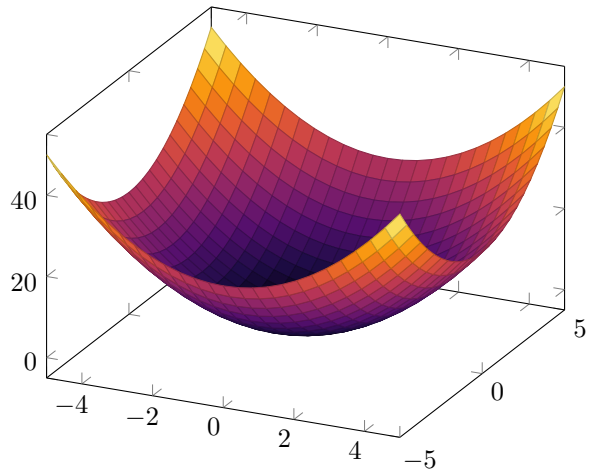
Also the first figure rotate x last, so even if the rotation angle are the same for y and we just said that rotating around x does nothing in this case (because the line is perfectly parallel tp x), the composition order is y then x, that is why it gave a different result.



Another example:

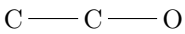


22.9 Surfaces - PgfPlots



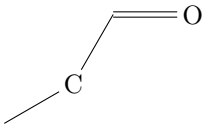
23 For chemists

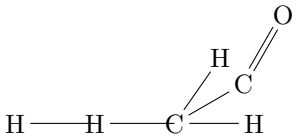
23.1 Simple links

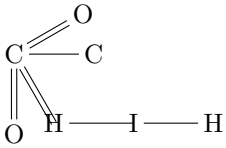


23.2 Links types

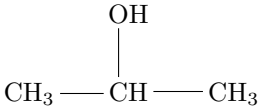
23.3 Links Angles





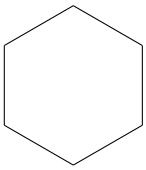


Isopropanol:

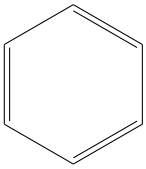


23.4 Rings

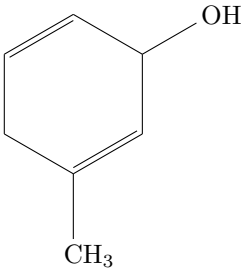
Cyclohexane:



Benzene:



Another example:



23.5 Charges

$N^{\oplus}$

$O^{\ominus}$

O^{-} O^{+}